

CNN (Convolutional Neural Network)

- for image dataset [Conv2D]
- for text dataset [Conv1D]
- for video dataset [Conv3D]

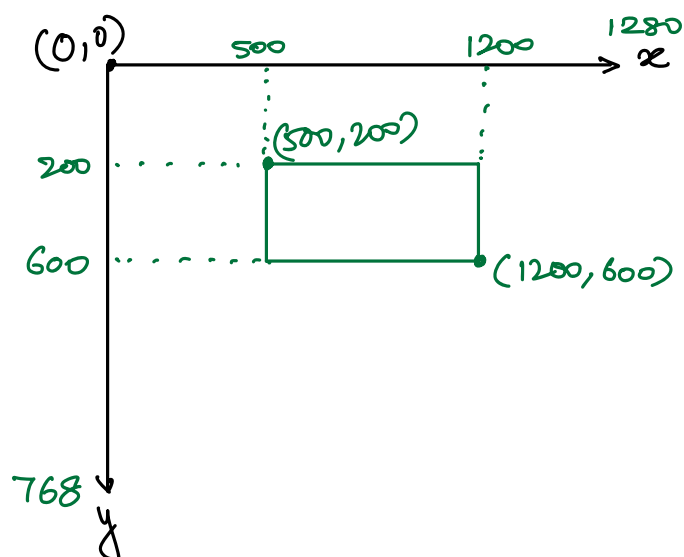
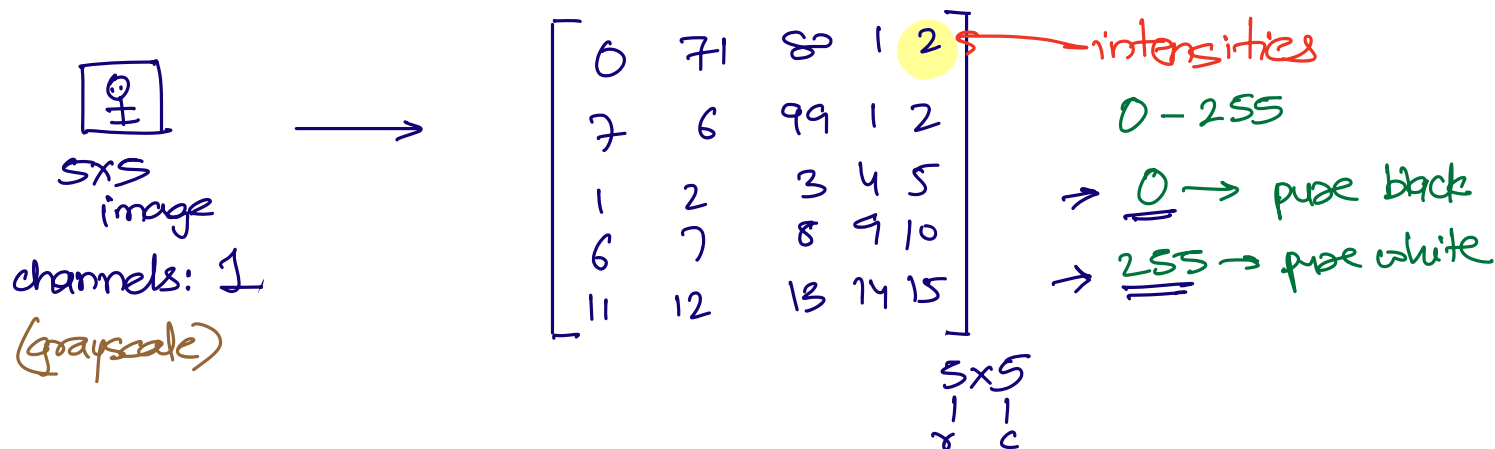
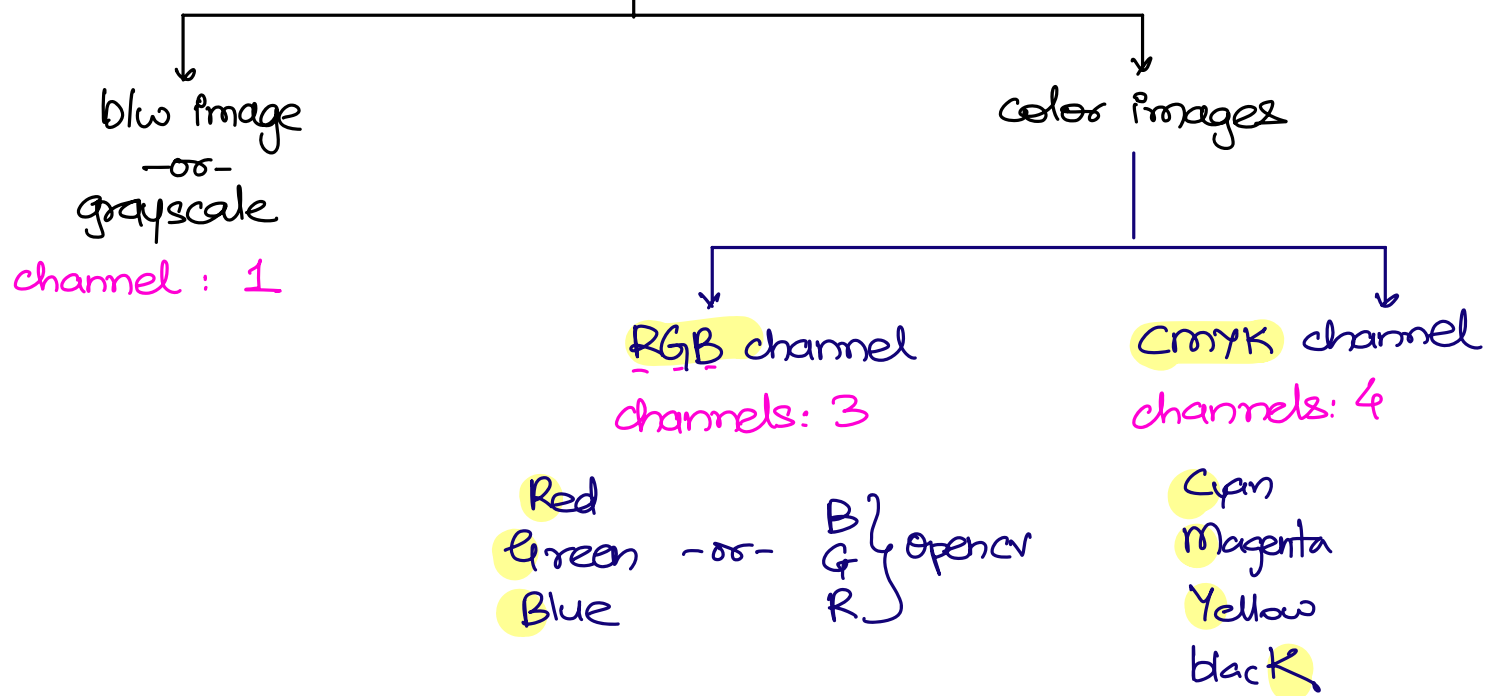


Image channels (represents the type of image)



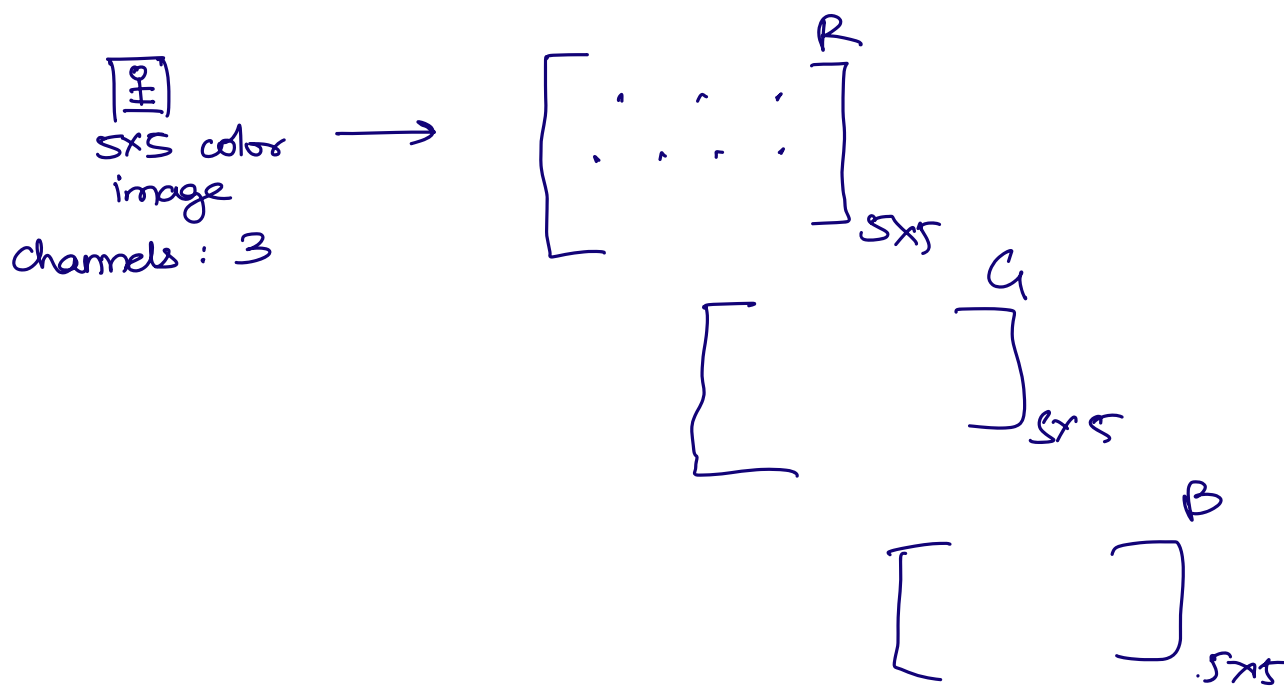
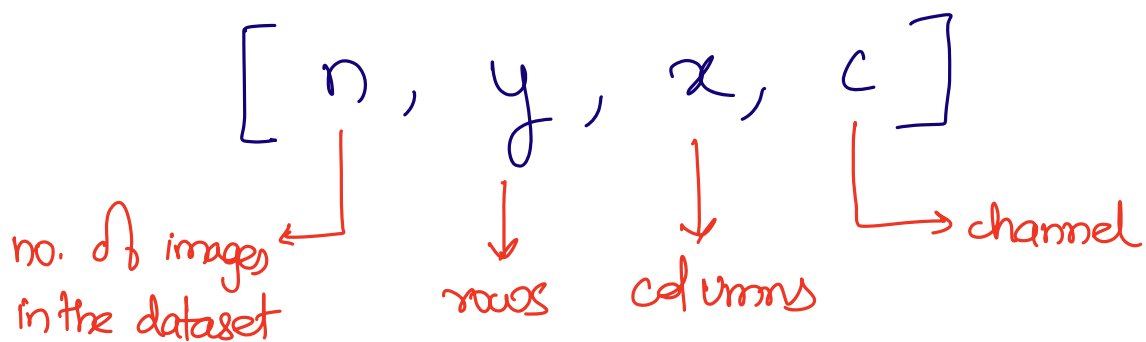


Image loading $\rightarrow$ Images are physically represented as



Fundamental Rules for preprocessing image dataset

① Always ensure the given images in your dataset are of SAME SIZE and is a square array.

$\underbrace{\hspace{10em}}$
 $\downarrow$
 no. of rows = no. of columns.

② Ensure all images in the given dataset follow SAME CHANNEL.

③ Always NORMALIZE the intensities of the images

$\frac{1}{255} \rightarrow$ all intensities will be in range of 0 & 1

Terminologies

① Kernel / Filter / Feature Detector

filter is used to extract features from an image

eg: 5x5 grayscale image

$$\begin{bmatrix} 3 & 3 & 2 & 1 & 0 \\ 0 & 0 & 1 & 3 & 1 \\ 3 & 1 & 2 & 2 & 3 \\ 2 & 0 & 0 & 2 & 2 \\ 2 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$\frac{1}{n} \times \frac{5}{y} \times \frac{5}{x} \times \frac{1}{c}$

image

⊗ convolve operation
(hadamard multiplication)

$$\begin{bmatrix} 0 & 1 & 2 \\ 2 & 2 & 0 \\ 0 & 1 & 2 \end{bmatrix}$$

filter ↑

→

$$\begin{bmatrix} 12 & 12 & 2 \\ 10 & 17 & 19 \\ 9 & 6 & 14 \end{bmatrix}$$

feature map.

3x3

② Strides

x, y shift of the filter

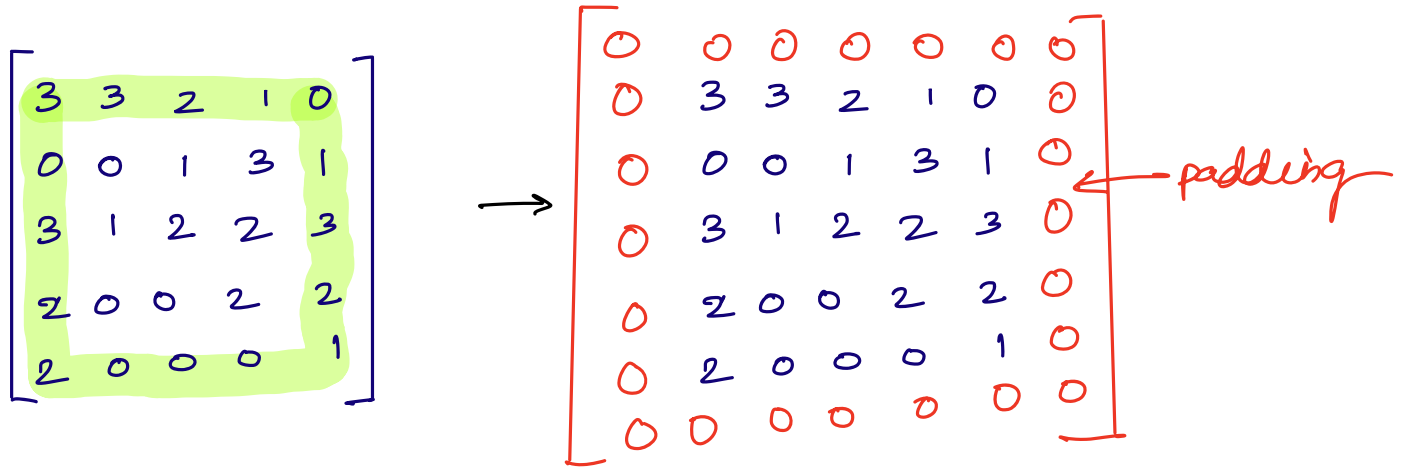
during convolve operation

default stride
config $\rightarrow \begin{matrix} x & y \\ (1, & 1) \end{matrix}$

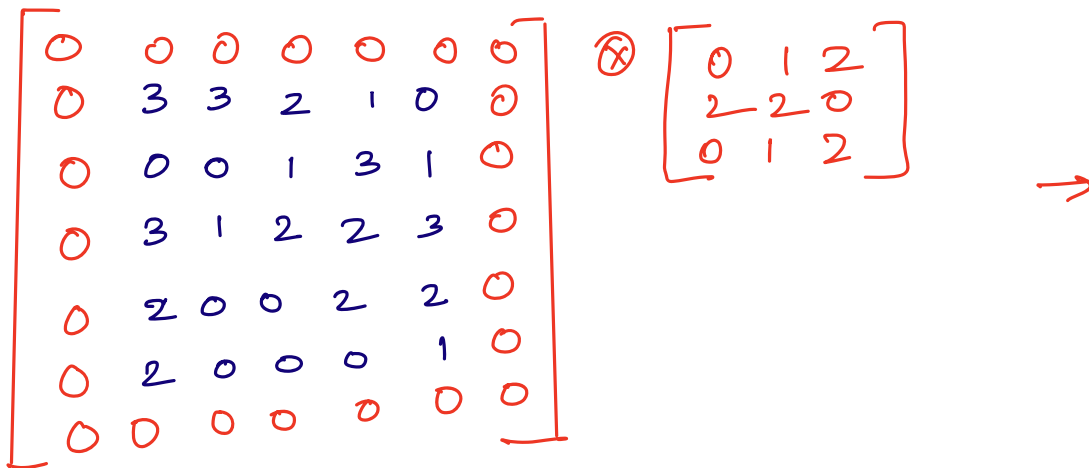
③ Padding

solution to address a UNIVERSAL COMPUTER GRAPHICS
problem called BORDER-EFFECT PROBLEM.

↳ Borders are always ignored.



Padding means adding VIRTUAL BORDER.



5x5

④ Pooling

used to downsample FEATURE MAPS

- or -

Feature Compression

Pooling is applied on feature Maps

Pooling Operations

Max Pooling
[take the max pixel from pool]

feature map

6	14	17	11	3
14	12	12	17	11
8	10	17	19	13
11	9	6	14	12
6	4	4	6	4

pool size
(2,2)

stride for
pooling = pool size

Average Pooling

apply max pooling

14	17
11	19

- feature maps

apply average pooling

11	15
10	15

$$\text{int} \left(\frac{6+14+14+12}{4} \right) = 11$$

⑤ flatten :

- applied on feature map

1	2	3
4	5	6
7	8	9

`ravel()`
[1, 2, 3, 4, 5,
6, 7, 8, 9]

FLATTEN
reshape it
to n,1
where
n → no of
elements in array

[1]
[2]
[3]
[4]
[5]
[6]
[7]
[8]
[9]